

Sensorimotor Proto-Objects: Bootstrapping Perception from Compensable Actions

Loïc Goasguen and Sylvain Argentieri

Abstract—Sensorimotor contingencies theory provides a strong foundation for a mathematical formalization of the bootstrapping problem of robot perception. Currently, this framework lacks the ability to deal with the dynamics of the environment and, by extension, objects. This paper shows that a sensorimotor agent with prediction functions acquired by motor babbling can infer environment dynamics and detect primitive object notions. To do that, we formalize Poincaré’s notion of compensation in terms of the agent’s action group, establishing an isomorphism with the environmental displacements. This allows the agent to describe environmental spatial transformations in terms of actions. We then conduct two experiments applying those developments to a visual task to (1) infer environmental global displacements and (2) infer a single object’s segmentation via its movement in the environment. We define a notion of sensorimotor proto-objects, as sensory elements whose information moves rigidly and coherently while maintaining shape and size. Here, the agent identifies the displacement by inferring the one that minimizes prediction error which is computed thanks to an internally built sensory metric. Results show that, in a simple 2D visual environment, the agent robustly infers the displacement observed for both the global and local case, but also accurately segments a single proto-object. By formalizing and inferring environmental transformation, this work introduces a new definition of sensorimotor proto-objects paving the way for a new understanding of how object perception can emerge from an agent’s own actions.

Keywords—sensorimotor contingencies, bootstrapping, embodied cognition, perceptual organization.

I. INTRODUCTION

IN his 1895 essay “*L’Espace et la Géométrie*”, Henri Poincaré argues that humans’ geometry stems from their ability to perform correlative actions to the solid bodies’ movements [1]. In other words, when an object translates in front of us, we are able to perform a displacement that puts us back in the same relative position to that object and thus retrieve the original sensory state. Here, geometry is the study of a specific (mathematical) group of displacement, *compensable transformations*. This perspective reflects that the relation between actions and sensations carries the structure of space. These arguments have found an echo in the sensorimotor

contingencies (SMC) theory of perception [2]. This theory portrays perception as a skillful activity: the active mastery of the sensorimotor regularities.

The work of Philipona *et al.* gathered the intuition of Poincaré and SMC theory into a mathematical framework, showing how a simple agent could deduce the dimensionality of the rigid group of spatial transformations through compensation [3]. This inspired the development of a sensorimotor approach to perception in robotics [4]. The core idea is to approach perception in an exclusively sensorimotor framework via a thorough mathematical formalization. In this perspective, the ability to perceive is no longer given *a priori* and must be actively learned through the sensorimotor flow. In turn, it allows the design of systems that can generalize robustly across diverse operational contexts.

The goal of formalizing mathematically is to derive formal proofs of the relations between the set of the agent’s knowledge and the structure ruling its interactions which imposes a careful understanding of the semantic assumptions underlying data processing [5]. It does so by detailing the constraints the environment imposes on the sensorimotor flow (i.e. from an external point of view), while simultaneously describing the agent’s naive access to experience (i.e., from an internal point of view). By articulating mathematically both points of view, this formalization illuminates the structure of sensorimotor contingencies. Unlike “black-box” approaches, the sensorimotor approach strives to make explicit the roots of the agent’s perceptive development while allowing the agent to learn solely from their experience with minimal preconceptions. In that sense, before building more autonomous robots, the sensorimotor approach to robotics aims to uncover the fundamental conditions of their autonomy.

Until recently, sensorimotor formalisms could not distinguish identical postures at different spatial locations, effectively restricting agents to stationary roles (e.g., [6], [7]). To address this, [8] introduced a formalism of *discrete* actions to distinguish between pose and posture. This allows an agent to compute sensorimotor relations between its displacements and their consequences via *prediction functions* that combine like actions, providing a means to capture the underlying structure of space. However, these implementations presuppose a static environment. Although extensions have addressed sensory noise [9], the inability to account for external dynamics remains a fundamental limit, preventing the framework from handling moving backgrounds or object interactions. In this paper, we propose a direct extension: formalizing how an agent’s mastery of its action structure enables it to infer environmental movements. Since objects cause such external displacements, identifying them allows segmenting the sensory array according

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to their motion. We thus introduce a primitive definition of a (rigid) object, or *proto-object*, based on the cohesive motion of its parts, providing a sensorimotor anchoring for discovering richer object concepts.

The remainder of this article is organized as follows. §II reviews the literature on object perception and displacement inference. §III provides the mathematical definitions for accessing external displacements. Based on this formalism, §IV presents experiments on capturing global transformations. These results are extended to local transformations in §V, formally defining the notion of proto-objects. Finally, §VI concludes the paper.

II. RELATED WORK

A. Objects and psychology

Gestalt psychology posits that object perception emerges from visual cues organized by simplicity and regularity [10]. Units similar in color, texture, or shape are grouped as unitary objects, a process mirrored in computer vision through explicit or learned feature extraction. However, SMC theory argues that object perception depends on lawful action-sensation relations [2] and must be mastered as a skill. Thus, static Gestalt cues alone cannot provide a sensorimotor perspective. Developmental work by Spelke [11] suggests that while some cues may be innate, others are acquired, noting that newborns rely primarily on the rigid, cohesive movement of elements for object perception [12].

This cohesive motion reflects the symmetry between perceiver and object movements found in compensatory transformations. Since these transformations are definable via sensorimotor regularities [3], [6], [13], we propose using them to form a primitive sensorimotor object notion upon which richer contingencies can be discovered. To implement this in robotics, we next review how agents infer displacements and objects.

B. Inferring displacements in robots

Visual odometry aims to infer robot displacements from visual field variations [14], while Visual Object Segmentation and Tracking (VOST) targets object identification. For instance, [15] leverages Spelke’s cohesive motion for segmentation. However, unsupervised VOST and odometry typically rely on high-level spatial priors or sensor geometry, while supervised VOST requires massive labeled datasets [16]. We argue that an unsupervised, sensorimotor definition of objects must bypass such assumptions, relying solely on the agent’s internal perspective.

This aligns with embodied approaches where interaction is the cornerstone of perception [17], closely linking SMCs to affordance learning [18]. Although some models leverage interaction for manipulation [19] or segmentation [20], they rarely treat interaction as consubstantial to perception, often requiring external *a priori* knowledge.

Conversely, bootstrapping approaches build knowledge from the ground up [5], [21]. While [22] defines an experiential object notion, it relies on sequentially learned high-level constructs like occupancy grids or sensory geometry [21].

We contend these are not prerequisites for the emergence of objecthood; inferring object displacements is as fundamental as perceiving the environment’s spatial structure.

Mastery of SMCs has been operationalized via probabilistic frameworks for object-following [23], categorization [24], or utility detection [25]. Shifting toward predictive accounts, [26] uses spatio-temporal invariance to detect proto-objects, aligning with SMCs viewed as action-oriented generative models [27]. Such models allow self-supervision by predicting the consequences of actions, a concept extending to active inference paradigms [28]–[30]. While active inference facilitates decision-making through prediction error minimization, its motivational aspects remain outside our current scope.

The following section formalizes how compensation allows an agent to relate external transformations to its own actions.

III. FORMALIZATION

A. A short recall on some definitions

The fundamental structure grounding the agent’s experience is space. As such, it has often been the subject of sensorimotor contingencies research, see [6], [13], [31]. Especially, a formalism making the structure of space transformations accessible through internal proxies of the agent’s actions is proposed in [8]. In all the following, we will heavily rely on the very same definitions which will be quickly recalled in the forthcoming paragraphs. The required mathematical notions are shortly recalled in the Supplementary Material.

1) *About the environment*: the ambient space is represented by the set $\mathcal{X}$ together with an automorphism group of admissible transformation $\mathcal{G}(\mathcal{X})$. Each element $\tau \in \mathcal{G}(\mathcal{X})$ corresponds to a rigid transformation, usually part of the special Euclidean group $SE^3(\mathbb{R})$. On this basis, a *state of the environment* $\epsilon \in \mathcal{E}$ is defined as a map $\epsilon : \mathcal{X} \rightarrow \mathcal{P}$ where $p = \epsilon(x) \in \mathcal{P}$ represents the value of the physical properties at position $x \in \mathcal{X}$, like color, temperature, luminance, etc.

2) *About the agent*: its embodiment is entirely described by its absolute motor configuration $\mathbf{b} = (\mathbf{m}, \tau) \in \mathcal{B}$, where $\mathbf{m} \in \mathcal{M}$ represents its internal motor configuration, and τ some absolute measure of its *pose* in the ambient space. The fact that the *pose* τ exploits the very same notation as the previous admissible transformations in space is not a coincidence. It allows to express that two absolute configurations are related to one another by a combination of translation and/or rotation when $\mathcal{G}(\mathcal{X})$ is $SE^3(\mathbb{R})$, i.e., a rigid transformation τ applied to some arbitrary absolute reference $\mathbf{b}_0$, see §2.3.1.2 in [32] for a detailed discussion.

Agents’ displacements are formalized by motor actions $a \in \mathcal{A}$ as maps $a : \mathbf{b} \rightarrow \mathbf{b}' = a\mathbf{b} = a(\mathbf{b})$. Importantly, we consider $\mathcal{A}$ as a group, meaning actions can be combined together, and every action a has an inverse a^{-1} such that $aa^{-1} = e$, the identity action. This definition of action is intrinsically “discrete” as it is defined as an instantaneous change in absolute motor configuration. However, this algebraic representation allows any action to be decomposed into a sequence of intermediate actions, thereby providing a formal basis for defining motor trajectories. The agent is also endowed with sensors. Those sensors are composed of sensory elements c_i called *sensels*

such that $\mathcal{C} = \{c_i\}$. Each sensel generates a sensation $s_i \in \mathcal{S}_i$ depending on the section of space it samples, which is captured by its receptive field $F_{c_i} : \mathcal{B} \rightarrow \mathcal{X}$. Thus, the sensation s_i can be written as $s_i = f_{c_i}(\epsilon|_{F_{c_i}(\mathbf{b})})$ where $f_{c_i} : \mathcal{P} \rightarrow \mathcal{S}_i$ denotes the sensel c_i sensitivity function capturing how physical properties in the environment –restricted to the receptive field of the sensel– are converted to sensations for the agent. This relation can be subsumed by the sensorimotor function ψ such that the sensory vector $\mathbf{s}$ for sensor $\mathcal{C}$ can be written as $\mathbf{s} = (s_i)_i = (\psi_i(\mathbf{b}, \epsilon))_i = \Psi(\mathbf{b}, \epsilon)$. The sensel definition is agnostic to the input nature, implying no strict requirement for raw pixel access. While higher-level features could be considered, this would introduce additional *a priori* assumptions.

3) *Discussion*: this formalism has been exploited in [8] to build the structure of the group of actions $\mathcal{A}$ by discovering the combinatorial properties of sensory prediction functions. In [8], these prediction functions take the specific form of permutations, later generalized in *partial* successor maps in [32]. But while these considerations allow to understand how the agent might structure its own set of actions, and to discover combinations or inverses of them, they are not explicitly related to the environment in which it is immersed. We hypothesize that predictable changes in an agent’s perception caused by its own actions can also serve as proxies for capturing environmental transformations. This is because certain displacements have the same sensory consequences, whether caused by the agent or by the environment. This idea, which links actions and transformations through the concept of compensation, is explored in the next subsection.

B. Sensorimotor compensation: towards transformations

Building on the formalism from the previous subsection, we elaborate a new definition of sensory compensation that incorporates actions. This allows us to formalize the theoretical links between an agent’s actions and environmental transformations.

1) *Definitions and properties*: let us consider an environment in a given state ϵ . One would characterize a (rigid) *movement* of this environment through a transformation $\tau \in \mathcal{G}(\mathcal{X})$ applied to this state, such that $\epsilon' = \tau\epsilon$ defines the new environment state after the transformation. Since the transformation is applied to the entire state ϵ , ϵ' is the same as ϵ but *shifted* in space. While not common in real-world situations, this is exactly the type of *global* transformations that Poincaré described to define its notion of *compensation* [1]. As such, the algebraic structure of compensation has been used to study space from a sensorimotor perspective in [3] or [13]. But differently from these works, we now propose to define compensation with the help of motor actions. Thus, an action a is said to *compensate* (or to be *compensable*) for sensory changes induced by a transformation $\tau \in \mathcal{G}(\mathcal{X})$ of the environment on every sensel of the set $\mathcal{C}$ if

$$\forall \mathbf{b} \in \mathcal{B}, \forall \epsilon \in \mathcal{E}, \Psi(\mathbf{b}, \epsilon) = \Psi(a\mathbf{b}, \tau\epsilon) = \mathbf{s}, \quad (1)$$

meaning that $\forall c_i \in \mathcal{C}, f_{c_i}(\epsilon|_{F_{c_i}(\mathbf{b})}) = f_{c_i}(\tau\epsilon|_{F_{c_i}(a\mathbf{b})}) = s_i$. Conversely, Eq. (1) defines a transformation τ as *compensable* if an action a exists that restores the original sensory state $\mathbf{s}$ after the transformation τ . This implies a conservation

rule for the agent’s initial sensation $\mathbf{s} = \Psi(\mathbf{b}, \epsilon)$. But this sensory conservation rule could also be written by considering the sensation $\mathbf{s}' = \Psi(\mathbf{b}, \tau\epsilon)$ obtained once the environment transformation is performed. To distinguish this case from the previous one, we say that an action a is *equivalent* for sensory changes induced by a transformation $\tau \in \mathcal{G}(\mathcal{X})$ of the environment on every sensel of the set $\mathcal{C}$ if

$$\forall \mathbf{b} \in \mathcal{B}, \forall \epsilon \in \mathcal{E}, \Psi(\mathbf{b}, \tau\epsilon) = \Psi(a\mathbf{b}, \epsilon) = \mathbf{s}', \quad (2)$$

meaning that $\forall c_i \in \mathcal{C}, f_{c_i}(\tau\epsilon|_{F_{c_i}(\mathbf{b})}) = f_{c_i}(\epsilon|_{F_{c_i}(a\mathbf{b})}) = s'_i$. Eq. (2) then states that performing either an equivalent action a on the current environment state or observing transformation τ of this environment leads to similar sensory variations. Furthermore, equivalent or compensable actions are related to each other: it is shown in Prop. S1 that if action a compensates τ then a^{-1} is equivalent to τ ; the same applies the other way around: if a is equivalent to τ , then a also compensates the inverse transformation $\tau^{-1} \in \mathcal{G}(\mathcal{X})$. We then propose to define the set of all compensable actions $\mathcal{A}_{\mathcal{G}(\mathcal{X})}$ without distinguishing them from equivalent actions, see Prop. S2. In the same vein, we define the set of all compensable or equivalent transformation $\mathcal{G}(\mathcal{X})_{\mathcal{A}}$ in Prop. S3. Importantly according to Prop. S3, $\mathcal{G}(\mathcal{X})_{\mathcal{A}}$ (resp. $\mathcal{A}_{\mathcal{G}(\mathcal{X})}$) is a subgroup of $\mathcal{G}(\mathcal{X})$ (resp. $\mathcal{A}$).

2) *Introducing predictable actions*: these definitions reveal that actions and their compensable transformations induce identical sensory variations. However, a formal link between the two still needs detailing. From the agent’s perspective, sensory variations can be captured by prediction functions $\Pi_a : \mathcal{S} \rightarrow \mathcal{S}$, with $\Pi_a \in \mathcal{F}(\mathcal{S}, \mathcal{S})$ [32]. An action a is then considered predictable if

$$\forall \mathbf{b} \in \mathcal{B}, \forall \epsilon \in \mathcal{E}, \exists \Pi_a \in \mathcal{F}(\mathcal{S}, \mathcal{S}), \Psi(a\mathbf{b}, \epsilon) = \Pi_a(\Psi(\mathbf{b}, \epsilon)). \quad (3)$$

Crucially, the predictability defined in Eq. (3) doesn’t imply an agent’s intrinsic predictive ability; rather, it simply states whether an action can be described by a deterministic function Π_a .

3) *Linking compensable actions and predictions*: from the definition of compensation in Eq. (1), a direct map between actions and their compensable transformations is not possible; a single action can compensate for various environment transformations, and vice versa. This is illustrated, for instance, when slightly different environment transformations are indiscernible to the agent’s sensor, or by actions enabling posture changes without any sensory apparatus movement. Consequently, the map $\Pi_{\mathcal{A}} : \mathcal{A}_{\mathcal{G}(\mathcal{X})} \rightarrow \Pi_{\mathcal{A}}(\mathcal{A}_{\mathcal{G}(\mathcal{X})})$ associating compensable actions to their sensory prediction function is non-injective.

However, given the agent’s minimal *a priori* knowledge, it can only discriminate actions by their distinct sensory consequences. Hence, let us consider the group of actions $\mathcal{A}$ and any subgroup of predictable actions $\mathcal{A}' \subseteq \mathcal{A}$. Godon

¹For propositions with the prefix ‘S’, please refer to the Supplementary Material.

²From now on we shorten “action a compensates for (or is equivalent to) sensory changes induced by transformation τ ” by “ a compensates τ ” and vice versa.

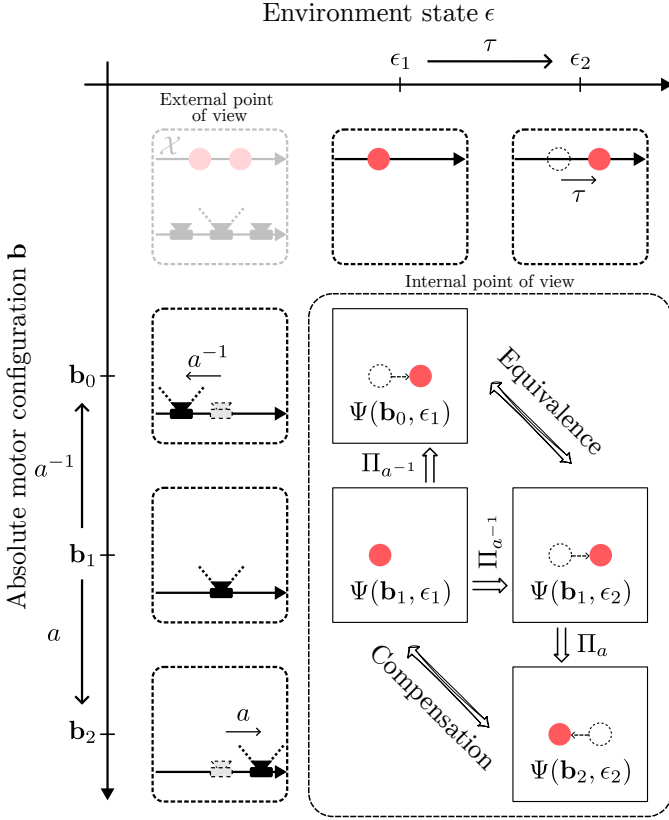


Figure 1. Illustration of compensable and equivalent actions. External point of view (top left): in a 1D space $\mathcal{X}$, an agent with a camera and a red disk environment translate along one axis. Internal point of view (bottom right): sensory states depend on environment transformation τ (horizontal axis), shifting from previous (dotted) to current states (red for disk, black for camera). Actions a or a^{-1} (vertical axis) either restore the original state (compensation) or reach the state induced by τ (equivalence). This shows how *external* transformations are captured by the same *internal* group structure as actions, linked via prediction functions Π_a .

shows that a map associating a subgroup of predictable actions $\mathcal{A}'$ to their prediction functions is a group homomorphism [32]. Since $\mathcal{A}_{\mathcal{G}(\mathcal{X})}$ is a group of predictable actions (by Prop. S3, S4), $\Pi_{\mathcal{A}}$ is a group homomorphism, and $\Pi_{\mathcal{A}}(\mathcal{A}_{\mathcal{G}(\mathcal{X})})$ is a group. A classical group theory result states that the kernel of a group homomorphism is a normal subgroup, inducing a group isomorphism [33]. For $\Pi_{\mathcal{A}}$, the kernel is $\ker \Pi_{\mathcal{A}} = \{a \in \mathcal{A}_{\mathcal{G}(\mathcal{X})} \mid \Pi_a(\Psi(\mathbf{b}, \epsilon)) = \Psi(\mathbf{b}, \epsilon)\}$, and the first isomorphism theorem yields

$$\mathcal{A}_{\mathcal{G}(\mathcal{X})} / \ker \Pi_{\mathcal{A}} \cong \Pi_{\mathcal{A}}(\mathcal{A}_{\mathcal{G}(\mathcal{X})}). \quad (4)$$

This means that each action³ of the quotient group maps to a single prediction function, and vice versa. Eq. (4) thus states that the structural properties of compensable actions are isomorphic to those of their respective prediction functions. This result mirrors findings in [8], extending them to any implementation of $\Pi_{\mathcal{A}}$ but restricting them to compensable actions only.

³Elements of a quotient group are cosets $a \ker \Pi_{\mathcal{A}}$, as they still refer to a set of sensory-indiscernible actions, we refer to each coset as an action a for clarity.

4) *Linking predictions and transformations*: let us continue the reasoning by adding transformations into the picture. It is shown in Corollary S1 that the prediction function associated with a compensable action can describe the sensory variation induced by its compensable transformation, thus establishing a formal link between predictions and transformations. Let us introduce the map $\Pi_{\mathcal{G}(\mathcal{X})} : \mathcal{G}(\mathcal{X})_{\mathcal{A}} \rightarrow \Pi_{\mathcal{A}}(\mathcal{A}_{\mathcal{G}(\mathcal{X})})$, which links any compensable transformation to the prediction function of its compensable action. By Prop. S5, $\Pi_{\mathcal{G}(\mathcal{X})}$ is a group homomorphism. Its kernel is defined as $\ker \Pi_{\mathcal{G}(\mathcal{X})} = \{\tau \in \mathcal{G}(\mathcal{X})_{\mathcal{A}} \mid a \text{ is equivalent to } \tau, \Pi_a(\Psi(\mathbf{b}, \epsilon)) = \Psi(\mathbf{b}, \epsilon)\}$. Consequently, according to the first isomorphism theorem,

$$\mathcal{G}(\mathcal{X})_{\mathcal{A}} / \ker \Pi_{\mathcal{G}(\mathcal{X})} \cong \Pi_{\mathcal{A}}(\mathcal{A}_{\mathcal{G}(\mathcal{X})}). \quad (5)$$

5) *Prediction as a proxy for compensable actions and transformations*: then, by composing Eq. (4) and (5), it immediately follows that

$$\mathcal{A}_{\mathcal{G}(\mathcal{X})} / \ker \Pi_{\mathcal{A}} \cong \Pi_{\mathcal{A}}(\mathcal{A}_{\mathcal{G}(\mathcal{X})}) \cong \mathcal{G}(\mathcal{X})_{\mathcal{A}} / \ker \Pi_{\mathcal{G}(\mathcal{X})}. \quad (6)$$

This establishes a formal link between compensable actions and transformations via prediction functions. Importantly, Eq. (6) demonstrates how *external* environment transformations can be captured by the same *internal* group structure as compensable actions, with their prediction functions binding the two. This key result shows that the agent can use its own actions, through predictions, as a proxy to understand environment transformations. This result is illustrated in Figure 1 for a very simple 1-dimensional interaction: in this example, prediction function Π_a associated with action a captures the same structure as the transformation τ .

IV. GLOBAL TRANSFORMATION INFERENCE

In this section, we exploit the demonstrated group isomorphism between transformations and predictions from Eq. (5) to infer global environment transformations. It is expected an agent can anticipate the consequences of its actions thanks to its prediction functions; we show in this section that it can also, *a posteriori*, identify environment transformations. This is achieved by minimizing the prediction error of hypothetical actions whose sensory prediction function maps to the observed transformation. To that end, we introduce a method based on the agent's ability to predict sensory outcomes.

A. Method

Inspired by predictive processing approaches [34] and action-oriented generative models [27], we propose a method for a naive agent to assess which external transformation is responsible for the changes in its sensory inputs when not directly stemming from its actions. In all the following, we will consider a discrete time setup, where n denotes the current timestep in the experiment. We consider the agent already built a set of prediction functions Π_a for its compensable actions $a \in \mathcal{A}_{\mathcal{G}(\mathcal{X})}$; the building process can be achieved via motor babbling as done in [8] and [9]. Consequently, it can anticipate the sensory consequences of its action a by computing the predicted sensory outcome $\hat{\mathbf{s}}^{(n+1)} = \Pi_a(\mathbf{s}^{(n)})$, where $\mathbf{s}^{(n)}$ is the sensory state at timestep n . If the agent

actually performs the action a concurrently with some global environment transformation τ , the actual sensory output $\mathbf{s}^{(n+1)}$ might differ from its prediction $\hat{\mathbf{s}}^{(n+1)}$. In order to infer the transformation τ , the agent then assesses which of its prediction functions best accounts for this discrepancy by minimizing the mean prediction error, as follows

$$\tilde{a} = \arg \min_{a_j} \frac{1}{|S|} \sum_{i=1}^{|S|} \text{PE}(s_i^{(n+1)}, \Pi_{a_j}(\hat{s}_i^{(n+1)})), \quad (7)$$

where PE is the function computing the prediction error between the actual sensory outcome $\mathbf{s}^{(n+1)}$, and the hypothesis applied to the expected sensory outcome $\hat{\mathbf{s}}^{(n+1)}$.

This hypothesis matching process resembles an *imagination* mechanism: the agent iterates through its known prediction functions to find the one that, when applied to its initial expected observation, minimizes the mean PE, thereby inferring the external transformation that occurred. This process implements a generative model where hypothesized sensory transitions are compared against the actual outcome $\mathbf{s}^{(n+1)}$. The inferred action produces sensory variations identical to the observed environmental discrepancy, making it equivalent to the transformation (Eq. (7)). While applying the prediction function to the observation $\mathbf{s}^{(n+1)}$ instead of the prediction $\hat{\mathbf{s}}^{(n+1)}$ would infer the inverse (compensatory) action, we prioritize the equivalent action as a direct application of the isomorphism in Eq. (6). Although directionality is often associated with the internalization of compensation, we argue that an explicit algorithmic mapping between compensatory actions and displacement directions is not required; the association emerges naturally from the interaction's structure. Fig. 2 illustrates this global inference method. To minimize *a priori* knowledge about the prediction error computation used in Eq. (7), we use the sensory distance d_S first introduced in [9] such that $\text{PE}(s_i, s_j) = d_S(s_i, s_j)$. This sensory distance is computed on the observed sensory symbol transition probabilities such that $d_S(s_i, s_j) = \mathbb{P}(s_j^{(n+1)} | s_i^{(n)})$, thus requiring no prior knowledge about the sensory space.

B. Experiments

1) *Setup*: to illustrate how actions can be used to infer transformations from Eq. (7), let us consider an agent navigating a 2D planar environment made of a static grayscale image. The agent, endowed with a 10×10 pixel camera, generates sensations by sampling this image. It can perform predefined actions drawn from the set $\mathcal{A} = \{a_{id}, a_l, a_r, a_f, a_b\}$, respectively corresponding to a null action and four cardinal movements, each altering the agent's sensory input by translating every sensel's receptive field by a single pixel width on the sensor. Crucially, the agent has no prior knowledge of these action spatial consequences. To simulate more realistic, noisy conditions, a centered Gaussian noise is added to the agent's sensory observations. This noise is tuned to achieve a constant Signal-to-Noise Ratio (SNR) for each observation while keeping the sensory values within the valid discrete range of 0 to 255.

Since Eq. (7) requires the agent to compare predicted and observed sensory outcomes for each action, the agent builds

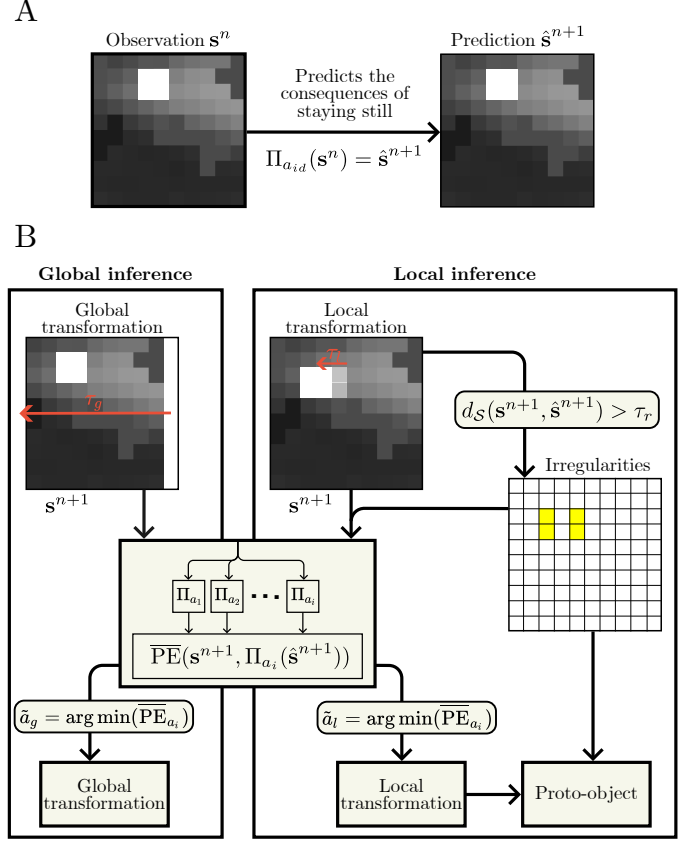


Figure 2. Illustration of the inference process for global and local environmental transformation for an agent performing the identity action. First, in A, upon choosing the action it will perform (here the identity action), the agent predicts its expected observation. In B, in the first case (left) the environments shifts globally from transformation τ_g . The agent then cycles through its prediction function assessing which one minimizes the mean prediction error between its prediction and the observation. In the second case (right) only part of the environment moves (the white square). According to its metric and a rejection threshold τ_r , it assess which sensels are irregular. Then it cycles through the prediction function on the irregularities to find the one that best minimizes the mean prediction error. Then it infer the external transformation and uses both that and the irregularities to segment the proto-object.

the sensory distance d_S from observing sensory transition probabilities between sensory symbols. To that end, the agent randomly draws actions from $\mathcal{A}$ and constructs the distance d_S based on the sensory transitions probabilities $\mathbb{P}(s_j^{(n+1)} | s_i^{(n)})$ caused by these actions. Then, prediction functions Π_a are built for each action $a_i \in \mathcal{A}$ as successor maps M_a , showing how sensory values are expected to shift when the agent performs action a_i . A successor map M_a is an $N \times N$ partial permutation matrix, where N is the number of sensels. For each sensel, it specifies which successor sensel will receive its sensory input. The partiality arises when an agent's sensory input previously outside of its receptive field is now in sight; a complete permutation map cannot describe such variation without referencing sensory values beyond the agent's current sensor state (see [8] for implementation details).

In this paper, the choice of a simplified 2D environment and a limited action set is a deliberate methodological step to provide a minimalist isolation of the sensorimotor mechanism and serves as an *existence proof* of the theoretical contribution.

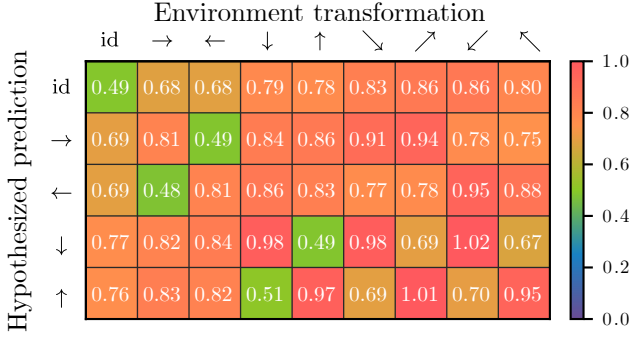


Figure 3. Mean prediction error $\overline{PE}$ between environment transformations τ_j (columns) and tested compensable actions a_i (rows).

While using raw pixel values and explicit permutation matrices yield limited scalability, with matrices becoming intractable with high dimensionality and non-repeatability of actions hindering exact predictions, it also ensures minimal *a priori* bias, demonstrating that the results are strictly caused by the internally discovered isomorphism.

Endowed with a sensory distance and prediction functions for its actions, the agent can now proceed with the action inference process from Eq. (7). During inference, the environment undergoes a global transformation τ_j in $\mathcal{G}(\mathcal{X})_A$, shifting the grayscale image of one pixel in the left, right, up, or down direction (the identity transformation is also possible). Note that these transformations are selected to exactly match compensable capabilities of the agent’s actions a_i in $\mathcal{A}_{\mathcal{G}(\mathcal{X})}$. Additionally, we also include diagonal environment transformations (top-left, top-right, bottom-left and bottom-right), which we know are not part of the agent’s compensable action set. For this experiment, the SNR is set to 40 dB, the agent performs 10,000 random actions drawn in $\mathcal{A}_{\mathcal{G}(\mathcal{X})}$. At each step, the environment can simultaneously undergo one of the nine aforementioned transformations.

2) *Assessing the prediction error*: we start by evaluating the mean prediction error $\overline{PE}$ of the agent’s actions relative to an external transformation τ_j . Inferring a transformation is expected to correlate with a minimal $\overline{PE}$. Fig. 3 shows that when the agent’s prediction function Π_{a_i} is sensory equivalent to the environment transformation τ_j , $\overline{PE}$ is significantly lower, allowing transformation inference. For non-compensable transformations (e.g., diagonal), the agent cannot internally represent them with its current action set. The results also illustrate how external transformations are internally captured: a global leftward environment shift is perceived as a rightward action, demonstrating how an agent’s perception of direction stems from the relationship between sensory information and its action set [35], as stated in the introduction. By changing the inference rule we could deduce the compensatory action instead of the equivalent one. Note that, given the demonstrated isomorphism in Eq. 6, the agent could extend its prediction functions to compute combinations of them and then correctly represent new transformations like the diagonal ones.

In summary, these results validate the sensorimotor agent’s ability to infer external global transformations. But given that global transformations are less common than moving

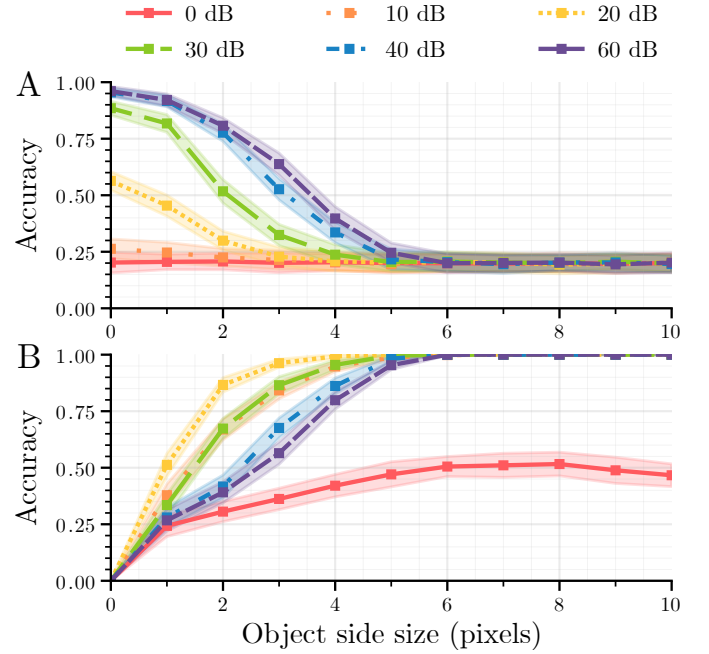


Figure 4. Mean (lines) and standard deviation (colored areas) of (A) environment and (B) object transformation inference accuracy versus object size l for various SNRs.

objects in real-world scenarios, the next section introduces an independently moving object.

3) *Inferring transformations with a moving object*: We evaluate simultaneous inference of global environment and independent object transformations, τ_j and τ_k , during action a_i . An opaque square of size $l \in [0, 10]$ is overlaid on the environment. At each step, the agent’s position is randomized, object appearance is regenerated from observed grayscale values, and the object’s center is reset to the sensor midpoint. Accuracy is measured over 10,000 steps across six SNRs (0, 10, 20, 30, 40, 60 dB). Fig. 4 shows environment accuracy (A) correlates with SNR (75%–100% at 40–60 dB for $l \leq 2$), but converges to random estimation (20%) as l increases. Object accuracy (B) is paradoxically favored by moderate noise, peaking at 20 dB SNR compared to 60 dB for $1 \leq l \leq 5$. Smooth background variations become indistinguishable from noise while steep object contrast remains salient, biasing the algorithm toward inferring object movement. At 0 dB SNR, extreme noise masks the object, reducing accuracy below 50%. Medium-sized objects cause the agent to select a non-explanatory “mean” action. The failure to resolve simultaneous transformations for extreme object sizes necessitates local transformation detection for precise object conceptualization.

V. LOCAL TRANSFORMATION INFERENCE

In environments typically explored by embodied agents, global transformations are essentially absent; only objects move independently or through interaction. Therefore, to infer external transformations, sensorimotor agents must identify and track these “local” external transformations—what one might call an object. In psychology, identifying objects and background is the problem of perceptual organization [10]. We

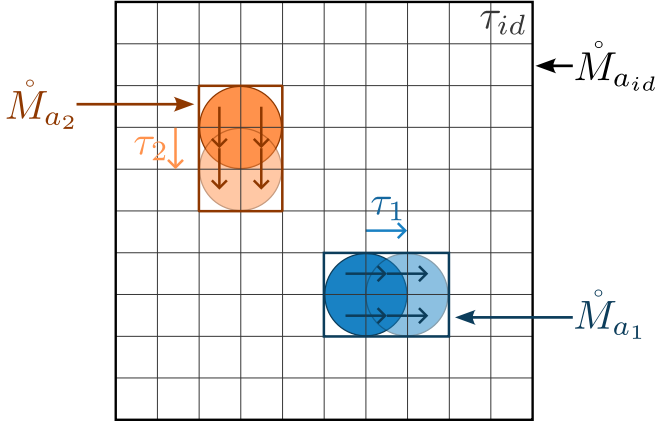


Figure 5. Illustration of the agent’s camera point of view. Two balls move according to two (sensory equivalent) transformations τ_1 and τ_2 . Their sensory variations can be described by local successor maps $\dot{M}_{a_1}$ and $\dot{M}_{a_2}$. Moreover the still background can also be described by τ_{id} and $\dot{M}_{a_{id}}$.

thus ground this problem in a sensorimotor paradigm by asking whether perceptual organization can be inferred via SMCs.

A. Introducing local successor maps

The formalization in §III was applied in the previous section to global transformations, captured through prediction functions Π_{a_i} linked to the agent’s compensable actions $a_i \in \mathcal{A}_{\mathcal{G}}(\mathcal{X})$. Our work utilizes prediction functions Π_{a_i} that describe sensory values shifts across sensels, taking the form of a successor map $\dot{M}_{a_i}$ introduced by [8], [9]. In §IV we applied prediction functions to the entire sensor observation, meaning Π_{a_i} is described by a *global* successor map. However, this concept can extend to a subset of sensels. Specifically, those the agent selects based on its ability to predict their outputs after its own action. This leads to the definition of a *local* successor map $\dot{M}_{a_i}$ of size $n_i \times n_i$, where n_i is the number of the agent’s predictable sensels after performing action a_i . Having $n_i = N$ (the total number of sensels) makes $\dot{M}_{a_i} = \dot{M}_{a_i}$. The necessity of describing local transformations is illustrated in Fig. 5, where the agent’s camera observes two balls undergoing different rigid transformations τ_1 and τ_2 while the environment remains static. As the previous section showed, the agent can’t effectively associate its global prediction functions Π_{a_i} with these two local transformations. However, if it can focus on the sensels where the moving balls produce sensory changes, it can then try to associate a local successor map $\dot{M}_{a_i}$ with each transformation.

Building on this intuition, we extend the previous formalization by introducing a new *total* successor map $M_{\mathcal{E}}$. This map aggregates local successor maps $\dot{M}_{a_i}$ of dimension $n_i \times n_i$ as follows

$$M_{\mathcal{E}} = \bigoplus_{i=0}^{N_{\mathcal{E}}-1} \dot{M}_{a_i}, \quad (8)$$

where $\bigoplus$ is the direct sum of matrices, and $N_{\mathcal{E}} = \sum_{i=0}^{T-1} n_i$ with T the total number of transformations detected in the environment. Taking Fig. 5 as an example, we would have $T = 3$, $n_0 = 88$ for the static environment, $n_1 = 6$ for the

first ball, $n_2 = 6$ for the second ball and the sensor has a size of $10 \times 10 = 100$ sensels. Then the *total* successor map $M_{\mathcal{E}}$ can be written as

$$M_{\mathcal{E}} = \begin{pmatrix} \dot{M}_{a_{id}} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \dot{M}_{a_1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \dot{M}_{a_2} \end{pmatrix}, \quad (9)$$

where a_1 and a_2 are the compensable actions associated with transformations τ_1 and τ_2 . Note that $M_{\mathcal{E}}$ is a block diagonal matrix. However, this disentangled form is a particular case, as the agent doesn’t *a priori* know how to order its sensels. Still, one could reorganize the map by permuting lines and columns to achieve such a description.

The agent might also move using some action a concurrently with local transformations, meaning sensory changes are caused by a combination of the agent’s action a and these local transformations. We can then define the *final* successor map $\overline{M}_{\mathcal{E}}$ as

$$\overline{M}_{\mathcal{E}} = M_a M_{\mathcal{E}} \quad (10)$$

where M_a describes the global successor map induced by the agent’s action a . Note that $N_{\mathcal{E}} = N$, as parts of the receptive field that are not describable by a prediction function are filled with zeros to match the total number of sensels, indicating that these sensels have no successors. While we don’t aim to prove that the relation of Eq. (6) holds for the *final* successor map $\overline{M}_{\mathcal{E}}$, we postulate that its underlying intuition still holds. That is, we consider $\overline{M}_{\mathcal{E}}$ a relevant representation of local transformations τ_i with their corresponding compensable actions a_i . This formalization allows us to define proto-objects as sets of sensels moving as coherent wholes. Let $\dot{M} = \{\dot{M}_{a_1}, \dots, \dot{M}_{a_{N_{\mathcal{E}}-1}}\}$ be the set of local successor maps within the total successor map $M_{\mathcal{E}}$. Then, at timestep $n + 1$, we define proto-object $\mathcal{O}_k^{(n+1)}$ as

$$\mathcal{O}_k^{(n+1)} = \left\{ c_j \in \mathcal{C} \mid \begin{array}{l} \exists c_i \in \mathcal{C}, \exists \dot{M}_{a_k} \in \dot{M} \text{ such that} \\ M_{\mathcal{E}}(i, j) = 1 \text{ and } \Pi_{a_k}(s_i^{(n)}) = s_j^{(n+1)} \end{array} \right\}. \quad (11)$$

Here, Π_{a_k} is the prediction function associated with the local successor map $\dot{M}_{a_k}$. We term these entities proto-objects to emphasize both their limited scope and their fundamental role in the emergence of the object concept. While the proto-object definition is external—known to the agent only as a byproduct of explaining environment transformations—its nature is inherently subjective and instantaneous. Defined as a set of sensels that experienced a predictable variation via the action prediction function, the proto-object is a product of the agent’s embodiment, not an objective external reality (e.g., one defined by spatial coordinates). Furthermore, its definition is strictly instantaneous, resulting only from the explicable consequences observed after a single timestep. Also, proto-object $\mathcal{O}_k^{(n+1)}$ relies on successor maps, and while this simplified approach aids theoretical development, it doesn’t scale to real robots with large sensors or complex actions. These successor maps currently describe only rigid, coherently moving objects that maintain shape, size, and move separately. These limitations are inherent to a developmental sensorimotor approach. As newborn object detection based on cohesive motion

precedes more complex Gestalt criteria [12], we postulate that aggregating elements through similar transformations provides a foundational assessment of space perception [1]. This initial segmentation based on rigid movement allows the subsequent discovery of richer properties, such as contiguity, contours, or shared textures.

However, for this intuition to hold, an agent must be able to actually detect the sensory consequences of local transformations τ_i within its entire sensory vector $\mathbf{s}$. Let us focus on this point in the next subsection.

B. Detecting local transformations: sensory segmentation

As discussed in §IV-B3, directly minimizing a *global* prediction error isn't effective for object identification; background transformations often dominate due to their larger error contribution. To allow the agent to focus on other relevant sensory changes, we introduce a minimal saliency mechanism, rooted in minimal *a priori* knowledge. We use the sensory distance d_S , quantifying transition frequencies to identify irregularities [9]. This assumes that unexpected sensory changes are more salient than regular ones, aligning with predictive coding frameworks where prediction errors drive attentional allocation [36]. By detecting specific sensels associated with these irregular transitions, the agent isolates relevant environmental changes for local inference. Accordingly, a sensel c_i is considered irregular if its predicted output value $\hat{s}_i^{(n+1)}$ deviates from the actual subsequent value $s_i^{(n+1)}$ by more than a fixed regularity threshold τ_r .

1) *Selecting the best hypothesis*: as in the global case, before each timestep the agent predicts its future sensory state $\hat{\mathbf{s}}^{(n+1)}$ using its prediction function Π_a for the action a it will perform. The set of irregular sensels at timestep $n+1$ can then be defined as

$$I^{(n+1)} = \{c_i \mid d(\hat{s}_i^{(n+1)}, s_i^{(n+1)}) > \tau_r\}, \quad (12)$$

with $\hat{s}_i^{(n+1)} = \Pi_a(s_i^{(n)})$. For each candidate transformation (with sensory equivalent action a_j), we compute the mean prediction error over the irregular set $I^{(n+1)}$ and select the one that minimizes the average prediction error as

$$\tilde{a}^{(n+1)} = \arg \min_{a_j} \frac{1}{|I^{(n+1)}|} \sum_{c \in I^{(n+1)}} \text{PE}(s_c^{(n+1)}, \Pi_{a_j}(\hat{s}_c^{(n+1)})). \quad (13)$$

This local version of the previous global inference provides a candidate action for the object's displacement. However, due to redundancy in the object, the set of detected irregularities does not always correspond to the actual object's position. The displacement of a pixel on a sensel previously occupied by the background can cause an irregularity concurrently with a sensel previously occupied by an object where we now see the background, thus irregularities are ambiguous.

2) *Multi-step refinement with objecthood probability tracking*: to resolve ambiguities, we introduce a temporal consolidation mechanism. When an action $\tilde{a}$ explains a subset of irregularities, we leverage the associated successor map $M_{\tilde{a}}$ to assess which sensels may be the cause of the irregularity and to instantiate the proto-object $\mathcal{O}$, in the sense of Eq. (11). This means that each sensel identified as a potential cause

of the irregularity, $c_i \in I^{(n+1)}$, is assigned to a proto-object perceptual unit $o_i \in \mathcal{O}$. This unit stores the identity of the sensel c_i and is assigned an initial *objecthood probability* score $P_{o_i}^{(0)}$. Each perceptual unit o_i is tracked over time: at each step, the sensel the unit is expected to be on is updated according to the successor map of the inferred transformation. If the new observation at the expected sensel matches the prediction, we reinforce its objecthood probability $P_{o_i}^{(n+1)}$ at timestep $n+1$ according to

$$P_{o_i}^{(n+1)} = \alpha \cdot P_{o_i}^{(n)} + (1 - \alpha) \cdot \delta, \quad (14)$$

where $\delta = 1$ for a match and $\delta = 0$ otherwise, and α a smoothing parameter ($0 < \alpha < 1$). Elements whose probability drops below a fixed threshold are discarded: consequently, the proto-object can persist, grow, or fade out depending on temporal consistency.

3) *Metrics*: to rigorously assess our segmentation, we introduce three key metrics based on the *objecthood probability* P_{o_i} of each detected unit $o_i \in \mathcal{O}$.

a) *Weighted Recall (WR)*: this metric quantifies the proportion of ground truth object elements that are successfully detected. It is calculated by summing the objecthood probabilities of detected proto-object units that belong to the ground truth set $\mathcal{O}^*$, normalized by the total size of the ground truth object (number of object pixels L):

$$\text{Weighted Recall (WR)} = \frac{\sum_{o_i \in (\mathcal{O} \cap \mathcal{O}^*)} P_{o_i}}{L}, \quad (15)$$

where L is the number of pixels of the ground truth object ($L = l \times l$ in our setup).

b) *Weighted Extended Precision (WEP)*: due to the ambiguity of sensels directly adjacent to an object's border in our minimal model, we define an extended ground truth set $\mathcal{O}_{\text{ext}}^*$ which includes the sensels belonging to $\mathcal{O}^*$ plus all directly object adjacent sensels. WEP quantifies the proportion of detected units that belong to this extended ground truth, thereby tolerating boundary errors. WEP is defined as:

$$\text{Weighted Extended Precision (WEP)} = \frac{\sum_{o_i \in (\mathcal{O} \cap \mathcal{O}_{\text{ext}}^*)} P_{o_i}}{\sum_{o_i \in \mathcal{O}} P_{o_i}}, \quad (16)$$

where the denominator sums the objecthood probabilities of all detected proto-object units.

c) *Weighted Mixed F1 (WMF1)*: this is the harmonic mean of WR and WEP, balancing detection completeness (WR) with border-tolerant accuracy (WEP):

$$\text{Weighted Mixed F1 (WMF1)} = \frac{2 \cdot \text{WEP} \cdot \text{WR}}{\text{WEP} + \text{WR}}. \quad (17)$$

A high WMF1 score indicates strong detection completeness (high WR) alongside minimized errors outside of the object's border (high WEP). We also compute the accuracy of the inferred transformation.

C. Experiment: Inferring the presence of one proto-object

1) *Experimental Setup*: the aim of the experiment is to demonstrate the ability to extend the previous method to the detection of one proto-object. For this, we'll use a minimal

2D setup, similar to our previous experiments. The agent’s camera is a 10×10 sensel grayscale sensor. Observations are subjected to the same noise process as before, with noise parameters adjusted relative to the observation power. To remain consistent with our minimal *a priori* framework, we compute the sensory metric for each SNR condition without any object present. This prevents biasing the experiment, as a small sensor and frequent object movements could incorrectly categorize object movements as regular. In a more realistic scenario, we hypothesize that object movements would appear irregular relative to other information (see [9] for metric construction experiments on real-world data, where most observations don’t involve constant object movement). For the metric building process, each noise condition (30, 40, and 60 dB) is computed over 10,000 successions of a 20×20 pixel sensor to increase metric construction speed. For the background, we use a black and white photograph of a house with large windows revealing nature. We place an object that can move in the up, down, right, and left directions with a movement amplitude causing a one pixel shift on the sensor. Its shape will always be a square of side size l . In this experiment, the object’s appearance is also randomly generated. The pixel values of the object are selected among observed sensory values during the sensory distance construction process. As each pixel is randomly generated, it is more likely that adjacent pixels of the object have a high sensory distance, making their displacement salient.

2) *Experimental process*: at each timestep, the agent remains at rest while the object moves via a transformation chosen randomly from those equivalent to $\{a_{id}, a_l, a_r, a_f, a_b\}$. The object cannot move within one pixel of the sensor border, ensuring constant visibility (as we are not assessing occlusion or object persistence). If one or more transformations are impossible due to this constraint, the object randomly selects from its remaining possible options. For each SNR value, we repeat the inference process for object sizes of 1×1 , 2×2 , 4×4 , and 6×6 pixels. Each inference scenario is assigned a random seed, influencing the initial position of the sensor on the background, the object’s relative position to the sensor, as well as the object’s appearance and trajectory. We perform 200 consecutive steps for each scenario, with 100 seeds per scenario. Experiments showed that setting the objecthood probability threshold, that discards a proto-object unit as part of the proto-object, to the same as the initial probability value improved results by directly discarding discernible outliers. Thus, we fix the initial probability and rejection threshold to 0.2. The object’s smoothing parameter α in Eq. 14 is set to 0.8; while this does not impact the final result, it facilitates visualizing the learning process compared to smaller rates. We also manually set the sensory distance regularity threshold to $\tau_r = 6$. The final process is summarized on the right part of Fig. 2.

3) *Results*: on Table I, we display the accuracy scores for the object’s transformation inference. We observe that the accuracy is positively correlated with SNR and is above 90% for 40 and 60 dB. Even in noisier conditions, accuracy stays above 85%. For each SNR, the 2×2 pixel object leads to the highest accuracy. For 40 dB and 60 dB it is the 1×1 pixel object that is systematically lower. For 30 and 40 dB the accuracy is inversely correlated with the object’s size (except for 1×1

Table I
TRANSFORMATION INFERENCE ACCURACY W.R.T. SNR AND OBJECT SIZE.

Object’s size	30 dB	40 dB	60 dB
1×1	0.903	0.915	0.925
2×2	0.967	0.999	1.000
4×4	0.895	0.990	1.000
6×6	0.859	0.961	1.000

objects). Once the object transformation is inferred, the agent forms a proto-object and associates its pixels with objecthood probability scores from Eq. (14). Fig. 6 shows the evolution of WMF1 scores for different object sizes. The score is maximum for 2×2 pixel objects and minimum for 1×1 pixel objects. While the score remains at 100% for a SNR of 60 dB, it decreases with object size at 30 and 40 dB. The score is also inversely correlated with noise, and the 1×1 case shows higher variance. To understand what causes the WMF1 scores, we observe the metrics used in its computation in Fig. 7. The figure shows WR and WEP for 40 dB SNR. WR, which stays between 0.85 and 1.0, is inversely proportional to object size, except for 1×1 pixel objects. WEP reaches 1 for every object size except for 1×1 pixel objects.

D. Discussion

The results show high efficiency in inferring local transformations of rigid objects and the ability to segment objects and their borders with minimal prior knowledge. The inference accuracies in Tab. I confirm that our attention mechanism successfully favors the contribution of objects over the background, a significant improvement over the large object results in Fig. 4. Furthermore, the WMF1 scores in Fig. 6 and the WR and WEP in Fig. 7 support that the sensels causing irregularities belong to the object. However, we observe lower scores for both small objects (1×1) and larger objects (above 2×2). An object’s ability to cause irregularities during its movement is crucial for detection, favoring objects with clearly defined edges. If an object is not big enough, it can easily go undetected if its pixels are regular compared to the background (as with 1×1 objects which have higher variance). Also, larger objects have a higher chance of displaying internal regularities, causing those sections to go undetected. In summary, our results demonstrate a high efficiency in inferring local transformations and segmenting objects and their borders using minimal prior knowledge.

We do not consider the low scores for 1×1 pixel objects a major issue, as this scale is rarely encountered in real-world robotics. The current experiment, conducted with a static background to specifically assess object segmentation, could be extended to a moving background; preliminary experiments show this is entirely possible, though it yields lower scores. The current method is also limited to single proto-object segmentation. In multi-object scenes, the agent must indeed untangle irregularities; however, its restricted number of actions and consequently the restricted number of possible object transformations makes it highly likely that irregularities will be miscategorized and fail to converge. Addressing multi-object segmentation necessitates extending

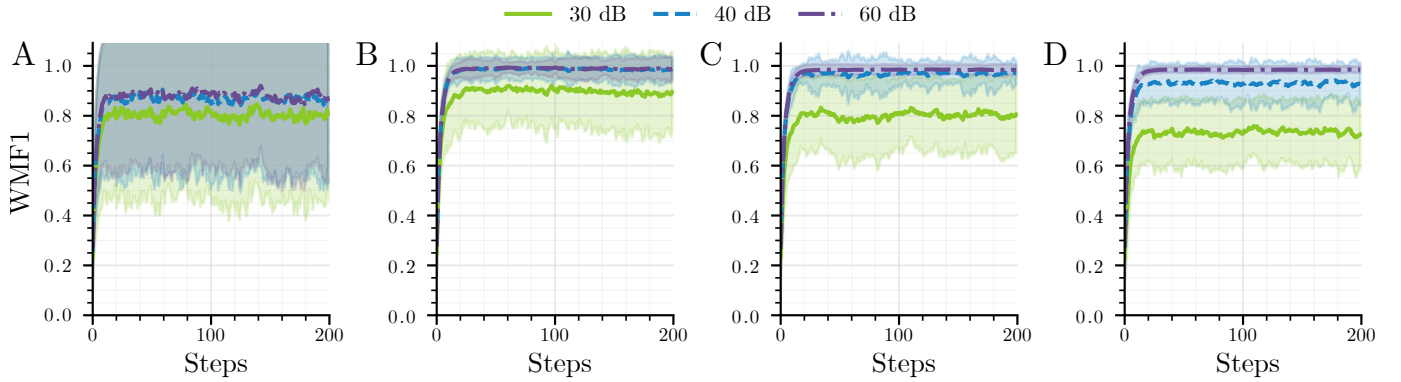


Figure 6. Evolution of mean and standard deviation of the WMF1 score relative to the SNR. Each figure depicts the WMF1 score for object sizes of 1-by-1 (A), 2-by-2 (B), 4-by-4 (C), and 6-by-6 (D) pixels. WMF1 has been obtained for SNR values of $\{30, 40, 60\}$ dB, respectively represented by the green, blue, and violet lines. Lines represent the mean and the shaded areas correspond to the standard deviation of the WMF1. Each curve represents the evolution of 100 different runs, each across 200 successive inference steps for a 10×10 pixel grayscale camera sensor.

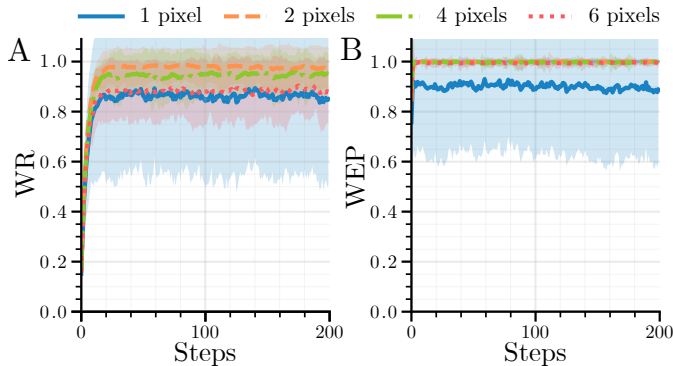


Figure 7. Evolution of the WR (A) and WEP (B) relative to the objects' sizes. Each parameter has been computed across 100 runs of 200 trials each with SNR equal to 40 dB. WR and WEP have been computed for 1×1 , 2×2 , 4×4 , and 6×6 pixel objects respectively represented by the blue, orange, green, and red lines. We represent the mean of those results by lines while shaded areas represent the standard deviation. The update rate is 0.8, the initial objecthood probability is 0.2, the objecthood probability rejection threshold is 0.2, and the sensory regularity threshold τ_r is 6.

the range of transformations the agent can infer and most likely the size of its sensor. Our method also relies on several *a priori* parameters, such as the update rate, regularity threshold τ_r , or initial objecthood probability. While our primary goal was structural clarity over efficiency, these parameters could be learned by using prediction error as a cost function, in line with predictive processing hypotheses [36]. The method's current difficulty in detecting proto-object units in regions where they appear regular, however, prevents it from scaling to larger, more realistic sensor sizes. This is partly due to the agent lacking a sensor geometry—a model of how sensors are arranged relative to each other in space. We could leverage the continuity of objects motion to identify neighboring sensors and thus create a sensor geometry (similar to [37]).

Our segmentation results do not try to rival state-of-the-art specialized methods [16]: the limited scalability of permutation matrices makes this implementation suited for the existence proof level of our theoretical contribution rather than serving as a real-world applicable method. Building on the idea that

identifying invariant sensorimotor structures is a prerequisite for object discovery [26], this work provides a foundational definition of proto-objects anchored exclusively within a group-theoretic framework. As a proof of concept, these results demonstrate that our formal framework can be implemented in a practical model without ambiguity, providing a robust basis for further advancements.

VI. CONCLUSION

In this paper, we addressed the problem of inferring environmental movement from SMCs for a naive robotic agent. We first used group theory to formalize the physical constraints that space imposes on both the agent and the environment. By establishing an isomorphism between compensable actions and external transformations, we showed that a single prediction function could describe both. This allowed us to develop a method for global environmental movement inference based on prediction errors from a naive sensory metric. The experiments demonstrated that while this approach was robust for global inference, it was insufficient for local displacement. We then introduced a saliency-based attention mechanism that enabled the agent to focus on irregularities, inferring the displacement of objects and segmenting them in 2D for small movements. We distinguish our theoretical sensorimotor perspective and generative architecture from the specific implementation using successor maps in raw pixel space. While the latter has limited scalability due to our focus on transparency and minimal *a priori* assumptions, we argue that the underlying formal framework could provide a robust and scalable foundation for future developments. The immediate limits to be addressed include extending the framework to handle more complex actions and object models, and scaling the algorithm for proper multi-object segmentation. To scale our approach, the concept of controllability can isolate salient elements from external noise [38]. Through competence acquisition, the agent grounds representations in learnable features affected by its own actions [39]. However, the current reliance on instantaneous actions precludes a full account of interaction dynamics. Transitioning to Lie groups and their associated algebras would allow for differentiable, infinitesimal actions, capturing

the continuous kinematics of sensorimotor flow through a more robust formal restructuring. Beyond these, our formal developments lay the foundation for more ambitious projects: developing a formal notion of a sensorimotor object. Key future directions include learning object interactions, defining features (like borders) as combinations of SMCs, and finding scalable prediction functions. Ultimately, such a model should also address other modalities to develop a multi-modal approach to sensorimotor objects.

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Supplementary Material to: “Sensorimotor Proto-Objects: Bootstrapping Perception from Compensable Actions”

Loïc Goasguen and Sylvain Argentieri

I. MATHEMATICAL PRELIMINARIES

The formalization used in the main text relies on a small set of elementary algebraic notions. For convenience, we briefly recall them here using the mathematical quantities introduced in the paper.

A. Maps and composition

A map $f : X \rightarrow Y$ associates to each element of a set X a unique element of a set Y . Several quantities used in the formalization of this paper are maps of this type:

- an environment state $\epsilon \in \mathcal{E}$ is a map $\epsilon : \mathcal{X} \rightarrow \mathcal{P}$;
- an action $a \in \mathcal{A}$ is a map $a : \mathcal{B} \rightarrow \mathcal{B}$, sending an absolute configuration $\mathbf{b}$ to a new one $a\mathbf{b}$;
- the sensorimotor map is written $\Psi : \mathcal{B} \times \mathcal{E} \rightarrow \mathcal{S}$;
- a sensory prediction function associated with an action is a map $\Pi_a : \mathcal{S} \rightarrow \mathcal{S}$.

B. Groups and subgroups

A group $(G, \cdot)$ is a set G equipped with an internal composition law satisfying four properties: closure, associativity, the existence of an identity element e , and the existence of an inverse g^{-1} for every $g \in G$. In the formalization, the set of actions $\mathcal{A}$ is treated as a group: composing two actions still gives an action, there exists an identity action leaving $\mathbf{b}$ unchanged, and each action a admits an inverse a^{-1} . Formally, $(\mathcal{A}, \cdot)$ is a group if

$$\left\{ \begin{array}{ll} \text{(closure)} & \forall a, a' \in \mathcal{A}, a' \cdot a \in \mathcal{A}, \\ \text{(associativity)} & \forall a, a', a'' \in \mathcal{A}, (a'' \cdot a') \cdot a = a'' \cdot (a' \cdot a), \\ \text{(identity)} & \exists e \in \mathcal{A}, \forall \mathbf{b} \in \mathcal{B}, e\mathbf{b} = \mathbf{b}, \\ \text{(inverse)} & \forall a \in \mathcal{A}, \exists a^{-1} \in \mathcal{A}, a^{-1} \cdot a = a \cdot a^{-1} = e. \end{array} \right. \quad (1)$$

In the paper, the composition law is simply denoted by juxtaposition, $a' \cdot a = a'a$. Likewise, the set of admissible rigid transformations $\mathcal{G}(\mathcal{X})$ is a group under composition, and its elements are denoted by τ .

A subset $H \subseteq G$ is a subgroup if it is itself a group for the same composition law. In the present setting, the sets of compensable actions $\mathcal{A}_{\mathcal{G}(\mathcal{X})} \subseteq \mathcal{A}$ and of compensable transformations $\mathcal{G}(\mathcal{X})_{\mathcal{A}} \subseteq \mathcal{G}(\mathcal{X})$ are the natural subgroup candidates studied below.

C. Homomorphisms, kernels, and quotient groups

A group homomorphism $\varphi : G \rightarrow H$ is a map that preserves composition: $\varphi(g_1 g_2) = \varphi(g_1) \varphi(g_2)$ for all $g_1, g_2 \in G$. In this work, the relevant examples are the maps that associate

compensable actions or compensable transformations with their sensory prediction functions.

The kernel of a homomorphism is the subset $\ker \varphi = \{g \in G \mid \varphi(g) = e_H\}$, where e_H is the identity element of the target group. It contains precisely the elements that become trivial through φ . For instance, $\ker \Pi_{\mathcal{A}}$ gathers the compensable actions whose induced prediction is the identity on sensory states.

Given a normal subgroup $N \triangleleft G$, one can define the quotient group G/N , whose elements are cosets gN . In the main formalization, quotient groups such as $\mathcal{A}_{\mathcal{G}(\mathcal{X})}/\ker \Pi_{\mathcal{A}}$ or $\mathcal{G}(\mathcal{X})_{\mathcal{A}}/\ker \Pi_{\mathcal{G}(\mathcal{X})}$ identify actions or transformations that have the same sensory prediction. The first isomorphism theorem then provides the link between these quotient groups and the corresponding prediction-function groups.

II. PROPOSITIONS AND COROLLARIES

This section gathers the propositions and corollary used in the main text, together with their proofs.

Proposition S1. Let a be a compensable action for a transformation τ . Then a^{-1} is an equivalent action for τ .

Proof. Select $a \in \mathcal{A}$ a compensable action for a transformation τ such that $\mathbf{b} = a^{-1}\mathbf{b}'$. Then

$$\begin{aligned} \forall \mathbf{b} \in \mathcal{B}, \forall \epsilon \in \mathcal{E}, \Psi(\mathbf{b}, \epsilon) &= \Psi(a^{-1}\mathbf{b}', \epsilon) = \Psi(a\mathbf{b}, \tau\epsilon) \\ &= \Psi(aa^{-1}\mathbf{b}', \tau\epsilon) \\ &= \Psi(\mathbf{b}', \tau\epsilon). \end{aligned} \quad (2)$$

Thus $\Psi(a^{-1}\mathbf{b}', \epsilon) = \Psi(\mathbf{b}', \tau\epsilon)$, showing that if action a compensates τ , then a^{-1} is equivalent to τ . $\square$

Proposition S2. Let $\mathcal{A}$ be the group of actions and $\mathcal{G}(\mathcal{X})$ the group of rigid transformations in space. Then the set of compensable actions $\mathcal{A}_C \subseteq \mathcal{A}$ coincides with the set of equivalent actions $\mathcal{A}_E \subseteq \mathcal{A}$, with

$$\begin{aligned} \mathcal{A}_C &= \{a \in \mathcal{A} \mid \exists \tau \in \mathcal{G}(\mathcal{X}), a \text{ compensates } \tau\}, \\ \mathcal{A}_E &= \{a \in \mathcal{A} \mid \exists \tau \in \mathcal{G}(\mathcal{X}), a \text{ is equivalent to } \tau\}. \end{aligned} \quad (3)$$

and we note $\mathcal{A}_{\mathcal{G}(\mathcal{X})} = \mathcal{A}_C = \mathcal{A}_E$ the common set of compensable and equivalent actions.

Proof. An action $a \in \mathcal{A}$ is defined as compensable, $a \in \mathcal{A}_C$, if for some $\tau \in \mathcal{G}(\mathcal{X})$, the relation $\Psi(\mathbf{b}, \epsilon) = \Psi(a\mathbf{b}, \tau\epsilon)$ holds for all states. Since τ induces a bijection on the environment states $\mathcal{E}$, we can substitute $\epsilon \rightarrow \tau^{-1}\epsilon$ to obtain an equivalent condition: $\Psi(\mathbf{b}, \tau^{-1}\epsilon) = \Psi(a\mathbf{b}, \epsilon)$. This is precisely the definition of an equivalent action, $a \in \mathcal{A}_E$, for the transformation $\tau' = \tau^{-1}$. Because $\mathcal{G}(\mathcal{X})$ is a group, an action is compensable for some τ if and only if it is equivalent for

its inverse τ^{-1} . Therefore, the sets of such actions must be identical, $\mathcal{A}_C = \mathcal{A}_E$, and is noted as $\mathcal{A}_{\mathcal{G}(\mathcal{X})}$. $\square$

Proposition S3. The set of compensatory actions $\mathcal{A}_{\mathcal{G}(\mathcal{X})}$ forms a subgroup of $\mathcal{A}$ and the set of compensable transformation

$$\mathcal{G}(\mathcal{X})_{\mathcal{A}} = \{\tau \in \mathcal{G}(\mathcal{X}) \mid \exists a \in \mathcal{A}, a \text{ compensates } \tau\} \quad (4)$$

forms a subgroup of $\mathcal{G}(\mathcal{X})$.

Proof. Identity elements $a_e \in \mathcal{A}$ and $\tau_e \in \mathcal{G}(\mathcal{X})$ satisfy $\Psi(\mathbf{b}, \epsilon) = \Psi(a_e \mathbf{b}, \tau_e \epsilon)$, ensuring that both identity elements are compensable. Let $a \in \mathcal{A}_{\mathcal{G}(\mathcal{X})}$ with its compensable transformation $\tau \in \mathcal{G}(\mathcal{X})_{\mathcal{A}}$, applying their inverses a^{-1} and τ^{-1} gives

$$\Psi(a^{-1} \mathbf{b}, \tau^{-1} \epsilon) = \Psi(\mathbf{b}, \epsilon),$$

proving closure under inversion. Let $a, a' \in \mathcal{A}_{\mathcal{G}(\mathcal{X})}$ with compensable transformations $\tau, \tau' \in \mathcal{G}(\mathcal{X})_{\mathcal{A}}$, we have

$$\Psi(a \mathbf{b}, \tau \epsilon) = \Psi(\mathbf{b}, \epsilon),$$

$$\Psi(aa' \mathbf{b}', \tau \tau' \epsilon') = \Psi(a' \mathbf{b}', \tau' \epsilon') \quad \text{using } \mathbf{b} = a' \mathbf{b}' \text{ and } \epsilon = \tau' \epsilon'$$

$$\Psi(aa' \mathbf{b}', \tau \tau' \epsilon') = \Psi(\mathbf{b}', \epsilon') \quad \text{by compensation}$$

proving closure under composition. Since $\mathcal{A}_{\mathcal{G}(\mathcal{X})}$ and $\mathcal{G}(\mathcal{X})_{\mathcal{A}}$ contain the identity, are closed under composition, and include inverses, they are subgroups of $\mathcal{A}$ and $\mathcal{G}(\mathcal{X})$, respectively. $\square$

Proposition S4. Any compensable action $a \in \mathcal{A}_{\mathcal{G}(\mathcal{X})}$ is predictable, we have

$$\forall \mathbf{b} \in \mathcal{B}, \forall \epsilon \in \mathcal{E}, a \in \mathcal{A}_{\mathcal{G}(\mathcal{X})}, \exists \Pi_a \in \mathcal{F}(\mathcal{S}, \mathcal{S}), \\ \Psi(a \mathbf{b}, \epsilon) = \Pi_a(\Psi(\mathbf{b}, \epsilon)),$$

or equivalently $\forall \epsilon \in \mathcal{E}, \forall \mathbf{b}, \mathbf{b}' \in \mathcal{B}, \Psi(\mathbf{b}', \epsilon) = \Psi(\mathbf{b}, \epsilon) \Leftrightarrow \Psi(a \mathbf{b}', \epsilon) = \Psi(a \mathbf{b}, \epsilon)$.

Proof. Let $\mathbf{b}, \mathbf{b}' \in \mathcal{B}$ be any absolute configurations such that $\forall \epsilon \in \mathcal{E}, \Psi(\mathbf{b}, \epsilon) = \Psi(\mathbf{b}', \epsilon)$. Since $a \in \mathcal{A}_{\mathcal{G}(\mathcal{X})}$ is compensable, there exists a transformation $\tau \in \mathcal{G}(\mathcal{X})_{\mathcal{A}}$ such that they are sensory equivalent, meaning that for all $\mathbf{b} \in \mathcal{B}$ and all $\epsilon \in \mathcal{E}$, $\Psi(a \mathbf{b}, \epsilon) = \Psi(\mathbf{b}, \tau \epsilon)$. Applying this equivalence to both $\mathbf{b}$ and $\mathbf{b}'$, we obtain

$$\Psi(a \mathbf{b}, \epsilon) = \Psi(\mathbf{b}, \tau \epsilon), \quad \text{and} \quad \Psi(a \mathbf{b}', \epsilon) = \Psi(\mathbf{b}', \tau \epsilon).$$

Since $\mathbf{b}$ and $\mathbf{b}'$ are assumed to be sensory equivalent to any ϵ , then $\Psi(\mathbf{b}, \tau \epsilon) = \Psi(\mathbf{b}', \tau \epsilon)$. As such for any $\epsilon \in \mathcal{E}$ we have

$$\forall \mathbf{b}, \mathbf{b}' \in \mathcal{B}, \Psi(a \mathbf{b}, \epsilon) = \Psi(a \mathbf{b}', \epsilon).$$

Thus for all $\epsilon \in \mathcal{E}$, we have $\Psi(a \mathbf{b}, \epsilon) = \Psi(a \mathbf{b}', \epsilon)$ proving that a satisfies the predictability condition. $\square$

Corollary S1. For any compensable action $a \in \mathcal{A}_{\mathcal{G}(\mathcal{X})}$ there exists $\tau \in \mathcal{G}(\mathcal{X})_{\mathcal{A}}$ such that

$$\forall \mathbf{b} \in \mathcal{B}, \forall \epsilon \in \mathcal{E}, \Pi_a(\Psi(\mathbf{b}, \epsilon)) = \Psi(\mathbf{b}, \tau \epsilon)$$

Proof. This follows directly from Proposition S4. By definition, a compensable action induces the same sensory consequences as the compensable transformation it addresses. Since Proposition S4 states that any compensable action is predictable (i.e., described by a deterministic function that models its sensory variation), it follows that this same prediction function

also describes the sensory variation induced by the compensable transformation. $\square$

Proposition S5. The map $\Pi_{\mathcal{G}(\mathcal{X})}$ that assigns a compensable transformations to an equivalent sensory prediction function, is a group homomorphism.

Proof. Let us consider groups $\mathcal{G}(\mathcal{X})_{\mathcal{A}}$ and $\Pi_{\mathcal{A}}(\mathcal{A}_{\mathcal{G}(\mathcal{X})})$. Let $\tau, \tau' \in \mathcal{G}(\mathcal{X})_{\mathcal{A}}$ and let $a, a' \in \mathcal{A}_{\mathcal{G}(\mathcal{X})}$ be any sensory equivalent actions associated with them, respectively. Since any compensatory action is also predictable (by Proposition S4), we have

$$\begin{aligned} \Psi(\mathbf{b}, \tau \tau' \epsilon) &= \Psi(aa' \mathbf{b}, \epsilon) = \Pi_{a'a}(\Psi(\mathbf{b}, \epsilon)) \quad (\text{predictability}) \\ &= \Pi_{a'}(\Pi_a(\Psi(\mathbf{b}, \epsilon))). \end{aligned}$$

Thus, $\Pi_{a'} \circ \Pi_a = \Pi_{a'a}$, confirming that $\Pi_{\mathcal{G}(\mathcal{X})}$ preserves composition, making it a group homomorphism. $\square$